

William Curran

PORTFOLIO: williamcurran.github.io

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SolidWorks Certified Associate | SEM (Scanning Electron Microscope) Trained & Certified

EDUCATION

University of Massachusetts Lowell — B.S., Mechanical Engineering

Expected May 2028

GPA: 3.5

- Relevant Coursework: Thermodynamics, Dynamics, Statics & Mechanics of Materials, Physics I II & III, Multivariable Calculus, Computer-Aided Design, Manufacturing Lab, Materials Science for Engineers

RESEARCH & LAB EXPERIENCE

Neural Electrode Fabricator (NEF) — Manufacturing Innovation and Neural Interface (MINI) Lab, Undergraduate Researcher

- Designed and built NEF, a precision laser-based system for processing ultra-fine neural microelectrode wire — multiple material types, each under 20 microns in diameter
- Capable of cutting wire tips to a sharp point and stripping insulation to an exact, user-specified length while leaving the internal conductor fully intact
- Built a Python-based data logging system that automatically records every test run to CSV, enabling fast, repeatable, data-driven process testing
- Used precision imaging and micromanipulation equipment for wire inspection and companion fabrication work

ROBOTICS PROJECTS

Industrial Robotic Arm

Spring 2026–Present

- Designed and built a 6-DOF robotic arm engineered for brute strength and repeatable motion: six NEMA 17 steppers, each driven through a custom 64:1 three-stage planetary gearbox (3D printed), giving every joint high torque and fine positional resolution
- Built a custom driver layout on a 12V motor supply, with an ESP32-S3 generating real-time step pulses for all six axes
- Mechanical and electrical builds are complete; currently developing PlatformIO firmware

Teleoperated Robotic Manipulator

Winter 2026

- Built a 4-DOF robotic arm that mirrors live human hand motion in real time: a Python/MediaPipe pipeline extracts hand landmarks and streams joint angles over serial to an Arduino driving MG996R servos
- Custom-designed and 3D-printed all structural links and mounts in SolidWorks; diagnosed insufficient holding torque and mechanical wobble under load, then resolved both by shortening arm length to cut joint moment and redesigning link geometry for rigidity

TECHNICAL SKILLS

- **Mechanical Design & Analysis:** SolidWorks (Certified Associate — advanced part/assembly modeling, GD&T), ANSYS (structural FEA)
- **Electrical & Embedded:** Custom control board design and assembly (soldering, perfboard/PCB layout), ESP32, TMC2209 stepper drivers, PlatformIO, closed-loop motor control, sensor/actuator integration from schematic to running hardware
- **Manufacturing:** Additive manufacturing — Formlabs SLA resin printing, FDM (PLA/PETG); CNC & lathe machining; precision laser micromachining
- **Software & AI-Assisted Development:** Python, computer vision (MediaPipe), AI/LLM-integrated coding and pipeline development, Flask/FastAPI